

SECTION 1: VLSM TREE

1.1 Base Network Derivation

Group size: 4 members. Rule: sum of last 3 digits of 4th member's ID.

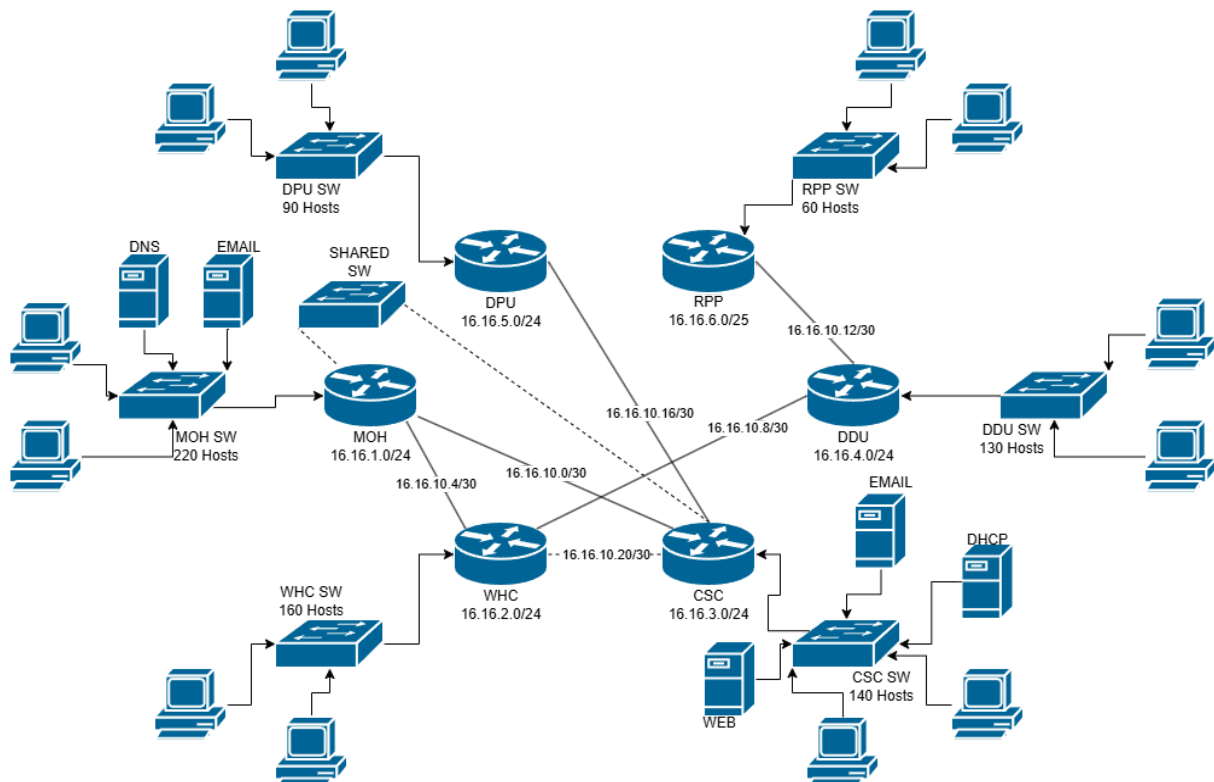
4th member ID: 22301349 --> Last 3 digits: 3, 4, 9 --> Sum = 16

Base Network: 16.16.0.0 / 16

1.2 Host Requirements per Center

Center	Devices	Required Hosts (with overhead)	Subnet Size	Prefix
MOH	220	226+	/24	254 hosts
WHC	160	165+	/24	254 hosts
CSC	140	148+	/24	254 hosts
DDU	130	134+	/24	254 hosts
DPU	90	94+	/24	254 hosts
RPP	60	64+	/25	126 hosts

1.3 VLSM Allocation Tree



SECTION 2: COMPLETE IP ADDRESS TABLE

2.1 MOH: Main Operations Hub (16.16.1.0/24)

Device	Interface	IP Address	Subnet Mask	Default Gateway
MOH Router	G0/0 (LAN)	16.16.1.1	255.255.255.0	N/A
MOH Router	S0/0/0 (->CSC)	16.16.10.1	255.255.255.252	N/A
MOH Router	S0/0/1 (->WHC)	16.16.10.5	255.255.255.252	N/A
PC1-MOH	NIC	16.16.1.10	255.255.255.0	16.16.1.1
PC2-MOH	NIC	16.16.1.11	255.255.255.0	16.16.1.1
DNS Server	NIC	16.16.1.2	255.255.255.0	16.16.1.1
Email Server	NIC	16.16.1.3	255.255.255.0	16.16.1.1

2.2 WHC: Warehouse Center (16.16.2.0/24)

Device	Interface	IP Address	Subnet Mask	Default Gateway
WHC Router	G0/0 (LAN)	16.16.2.1	255.255.255.0	N/A
WHC Router	S0/0/0 (->MOH)	16.16.10.6	255.255.255.252	N/A
WHC Router	S0/0/1 (->DDU)	16.16.10.9	255.255.255.252	N/A
WHC Router	S0/1/0 (->CSC bkp)	16.16.10.21	255.255.255.252	N/A
PC1-WHC	NIC	DHCP	255.255.255.0	16.16.2.1
PC2-WHC	NIC	DHCP	255.255.255.0	16.16.2.1

2.3 CSC: Customer Support Center (16.16.3.0/24)

Device	Interface	IP Address	Subnet Mask	Default Gateway
CSC Router	G0/0 (LAN)	16.16.3.1	255.255.255.0	N/A
CSC Router	S0/0/0 (->MOH)	16.16.10.2	255.255.255.252	N/A
CSC Router	S0/0/1 (->DPU)	16.16.10.17	255.255.255.252	N/A
CSC Router	S0/1/0 (->WHC bkp)	16.16.10.22	255.255.255.252	N/A
PC1-CSC	NIC	DHCP	255.255.255.0	16.16.3.1
PC2-CSC	NIC	DHCP	255.255.255.0	16.16.3.1
Web Server	NIC	16.16.3.2	255.255.255.0	16.16.3.1
Email Server	NIC	16.16.3.3	255.255.255.0	16.16.3.1
DHCP Server	NIC	16.16.3.4	255.255.255.0	16.16.3.1

2.4 DDU: Delivery Dispatch Unit (16.16.4.0/24)

Device	Interface	IP Address	Subnet Mask	Default Gateway
DDU Router	G0/0 (LAN)	16.16.4.1	255.255.255.0	N/A
DDU Router	S0/0/0 (->WHC)	16.16.10.10	255.255.255.252	N/A
DDU Router	S0/0/1 (->RPP)	16.16.10.13	255.255.255.252	N/A
PC1-DDU	NIC	DHCP	255.255.255.0	16.16.4.1
PC2-DDU	NIC	DHCP	255.255.255.0	16.16.4.1

2.5 DPU: Data Processing Unit (16.16.5.0/24)

Device	Interface	IP Address	Subnet Mask	Default Gateway
DPU Router	G0/0 (LAN)	16.16.5.1	255.255.255.0	N/A
DPU Router	S0/0/0 (->CSC)	16.16.10.18	255.255.255.252	N/A
PC1-DPU	NIC	16.16.5.10	255.255.255.0	16.16.5.1
PC2-DPU	NIC	16.16.5.11	255.255.255.0	16.16.5.1

2.6 RPP: Regional Pickup Point (16.16.6.0/25)

Device	Interface	IP Address	Subnet Mask	Default Gateway
RPP Router	G0/0 (LAN)	16.16.6.1	255.255.255.128	N/A
RPP Router	S0/0/0 (->DDU)	16.16.10.14	255.255.255.252	N/A
PC1-RPP	NIC	16.16.6.10	255.255.255.128	16.16.6.1
PC2-RPP	NIC	16.16.6.11	255.255.255.128	16.16.6.1

SECTION 3: ROUTER CONFIGURATION COMMANDS

MOH:

```
enable
configure terminal
hostname MOH
interface g0/0
ip address 16.16.1.1 255.255.255.0
no shutdown
interface s0/0/0
ip address 16.16.10.1 255.255.255.252
clock rate 64000
no shutdown
interface s0/0/1
ip address 16.16.10.5 255.255.255.252
clock rate 64000
no shutdown
exit
router rip
version 2
no auto-summary
network 16.16.1.0
network 16.16.10.0
passive-interface g0/0
redistribute static
ip route 16.16.4.0 255.255.255.0 16.16.10.6
end
write memory
```

WHC:

```
enable
configure terminal
hostname WHC
interface g0/0
ip address 16.16.2.1 255.255.255.0
no shutdown
interface s0/0/0
ip address 16.16.10.6 255.255.255.252
no shutdown
interface s0/0/1
ip address 16.16.10.9 255.255.255.252
clock rate 64000
no shutdown
interface s0/1/0
ip address 16.16.10.21 255.255.255.252
clock rate 64000
no shutdown
exit
ip dhcp excluded-address 16.16.2.1 16.16.2.10
ip dhcp pool WHC-LAN
network 16.16.2.0 255.255.255.0
default-router 16.16.2.1
dns-server 16.16.1.2
router rip
version 2
no auto-summary
network 16.16.2.0
network 16.16.10.0
passive-interface g0/0
ip route 16.16.3.0 255.255.255.0 16.16.10.22 121
end
write memory
```

DDU:

```
enable
configure terminal
hostname DDU
interface g0/0
ip address 16.16.4.1 255.255.255.0
no shutdown
interface s0/0/0
ip address 16.16.10.10 255.255.255.252
no shutdown
interface s0/0/1
ip address 16.16.10.13 255.255.255.252
```

```
clock rate 64000
no shutdown
exit
ip dhcp excluded-address 16.16.4.1 16.16.4.10
ip dhcp pool DDU-LAN
network 16.16.4.0 255.255.255.0
default-router 16.16.4.1
dns-server 16.16.1.2
router rip
version 2
no auto-summary
network 16.16.4.0
network 16.16.10.0
passive-interface g0/0
end
write memory
```

CSC:

```
enable
configure terminal
hostname CSC
interface g0/0
ip address 16.16.3.1 255.255.255.0
no shutdown
interface s0/0/0
ip address 16.16.10.2 255.255.255.252
no shutdown
interface s0/0/1
ip address 16.16.10.17 255.255.255.252
clock rate 64000
no shutdown
interface s0/1/0
ip address 16.16.10.22 255.255.255.252
no shutdown
exit
router rip
version 2
no auto-summary
network 16.16.3.0
network 16.16.10.0
passive-interface g0/0
ip route 16.16.6.0 255.255.255.128 16.16.10.1
interface g0/0
ip helper-address 16.16.3.4
end
write memory
```

DPU:

```
enable
configure terminal
hostname DPU
interface g0/0
ip address 16.16.5.1 255.255.255.0
no shutdown
interface s0/0/0
ip address 16.16.10.18 255.255.255.252
no shutdown
exit
ip route 16.16.6.0 255.255.255.128 16.16.10.17
ip route 16.16.1.0 255.255.255.0 16.16.10.17
ip route 16.16.2.0 255.255.255.0 16.16.10.17
ip route 16.16.3.0 255.255.255.0 16.16.10.17
ip route 16.16.4.0 255.255.255.0 16.16.10.17
end
write memory
```

RPP:

```
enable
configure terminal
hostname RPP
interface g0/0
ip address 16.16.6.1 255.255.255.128
no shutdown
interface s0/0/0
ip address 16.16.10.14 255.255.255.252
no shutdown
exit
ip route 16.16.1.0 255.255.255.0 s0/0/0
ip route 0.0.0.0 0.0.0.0 s0/0/0
end
write memory
```

CSC Dedicated DHCP Server

Field	Value
Pool Name	CSC_LAN
Default Gateway	16.16.3.1
DNS Server	16.16.1.2
Start IP Address	16.16.3.11
Subnet Mask	255.255.255.0
Max Users	243

SECTION 4: ASSUMPTIONS

A1	Floating Static AD = 121 RIPv2 has a default Administrative Distance of 120. The floating static route on WHC to CSC is assigned AD=121 so it only activates upon failure of the primary RIP path.
A2	MOH Shared Switch MOH and CSC are connected via a common shared switch as required. In Packet Tracer this is a separate switch connecting both routers' LAN interfaces, with PCs/servers on local switches.
A3	Email Relay Email servers in MOH and CSC are enabled with SMTP service. CSC's email server is designated as the relay for all centers. Full SMTP relay is beyond Packet Tracer's simulation scope.
A4	Static-only Routers (DPU, RPP) DPU and RPP do not run RIPv2 as explicitly stated in the requirements. They use only static routes. DPU uses CSC as sole gateway; RPP uses a default route pointing to DDU.